Wine Quality Prediction Using a Weighted Ensemble

Review 2 — class balancing, improving the result, and feature importance

Building on a reproduction of Ma et al. (2024), Journal of Business Research 176:114576

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Second Review

Where Review 1 left off

The baseline, the diagnosis, and the three changes requested

69.01%
REVIEW-1 ENSEMBLE (FAITHFUL REPRODUCTION)

71.03%
RANDOM FOREST ALONE — BEAT THE ENSEMBLE

0.6%
CLASS-4 RECALL — RARE CLASSES IGNORED

≈ 0.25
EVERY WEIGHT — TOO FLAT TO HELP

Diagnosis at Review 1: the paper's weights come out nearly uniform, so the ensemble behaves like a plain average — it lands between its members instead of above the best one.

Requested at Review 1	What Review 2 did	Slides
1. Class balancing — yet to be done	Four balancing schemes compared on identical folds, incl. SMOTE inside the fold	4 – 5
2. Improve the result	Tested the Review-1 diagnosis, then fixed it: weights tuned by inner CV; stacking; stronger members	6 – 10
3. Work on feature importance	Permutation importance on held-out folds, checked against two other methods and the paper	11 – 14

How Review 2 is judged

Rules fixed in the code before any Review-2 experiment was run

1

Same protocol as Review 1

Stratified 10-fold CV × 10 repeats, the same fold seeds, every scaler and resampler fitted inside the training fold.

2

Review 1 is not rewritten

Review-1 numbers are unchanged. Re-run inside the new code, the Review-1 setup reproduces 69.01% to every digit — so every comparison is like-for-like.

3

Primary metric: macro-F1

Averages F1 over all six quality classes with equal weight. Accuracy is still reported on every slide.

Why not just accuracy?

The Review-1 ensemble scores 69.0% accuracy while getting 0.0% of class-3 wines and 0.6% of class-4 wines right. Classes 5 and 6 are 82.5% of the data, so accuracy barely notices the rare grades. Class balancing deliberately trades a little of the common classes for the rare ones — judging it on accuracy alone would judge it on the one thing it gives up.

Disclosure: the winner is picked on the same cross-validation that scores it, which flatters its score slightly. Two candidates were added after a 1-repeat smoke test; their settings are chosen inside each training fold, never from the test folds. Both are stated rather than hidden.

Class balancing — four schemes, one variable

Same four learners, same paper weighting, same folds — only the balancing changes

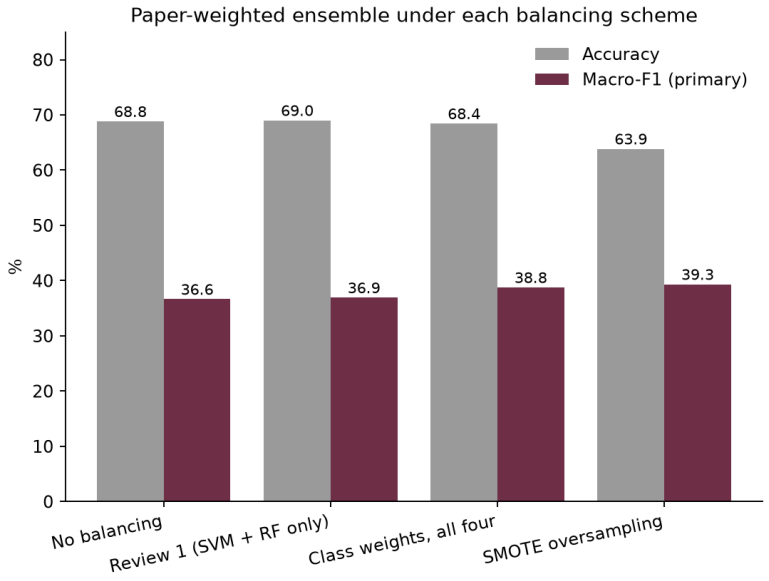
Scheme	What it does	Where it acts
No balancing	Every training sample counts equally — the rare classes are simply outvoted	—
Review 1	'Balanced' class weights on SVM and Random Forest only	Inside the loss
Class weights, all four	The same for MLP and GradBoost too. They have no class_weight option, so each sample gets the equivalent weight directly	Inside the loss
SMOTE oversampling	Creates synthetic rare-class wines by interpolating between real neighbouring ones, until every class is equally represented	Training fold only

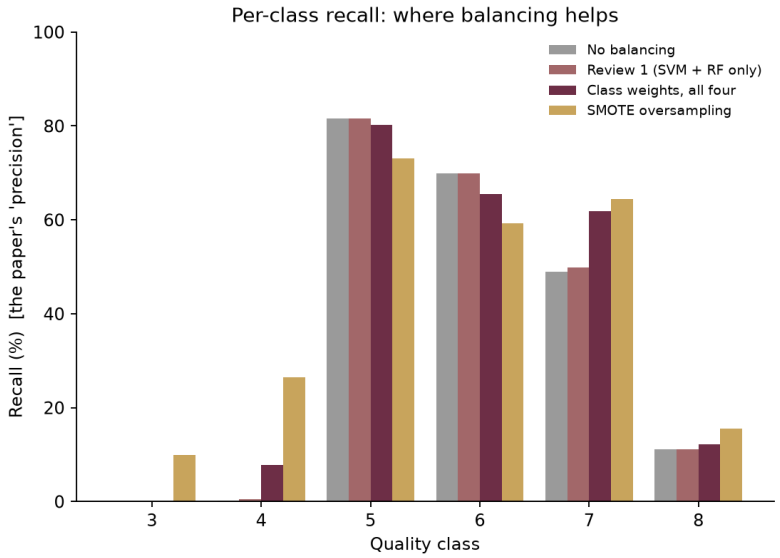
SMOTE without leakage

SMOTE sits inside the pipeline, after the scaler: training fold → scale → synthesise → fit. Synthetic wines are built only from training-fold wines, and the test fold is never resampled — the same rule that fixed the leakage bug at Review 1.

Class balancing — results

Paper-weighted ensemble, pooled over 100 fold-runs





What it shows

- SMOTE oversampling gives the best macro-F1: 39.34 vs 36.92 at Review 1
- Class-4 recall: 0.6% → 26.4% with SMOTE; class 7: 49.9% → 64.5%
- The price: accuracy falls 5.15 points, because the model now calls rare classes more often and some of those calls are wrong
- Balancing moves recall from the crowded middle classes to the rare ones — it does not create information that isn't there

Scheme	Acc.	Macro-F1	Rec. 4	Rec. 7	Rec. 8
No balancing	68.82	36.64	0.2	49.0	11.1
Review 1 (SVM + RF only)	69.01	36.92	0.6	49.9	11.1
Class weights, all four	68.43	38.77	7.7	61.9	12.2
SMOTE oversampling	**63.86**	**39.34**	**26.4**	**64.5**	**15.6**

Improving the result — four routes

Each one tests a specific idea; one of them is the fix the Review-1 diagnosis points to

1 TEST

Sharpen the weights

Raise the paper's weights to a power T before normalising. Review 1 predicted sharper weights would let the ensemble overtake Random Forest.

A diagnostic: the weights see the predictions they are scored on, so it is never used to pick the final model.

2 FIX

Learn the temperature

The same sharpening, but T is chosen inside each training fold by an inner 5-fold CV — the test fold is never involved.

The leak-free version of route 1: a legitimate result.

3 REPLACE

Stacking

Throw away the hand-made weights; a logistic regression learns how to combine the members' probabilities.

Also tried with a class-balanced combiner.

4 UPGRADE

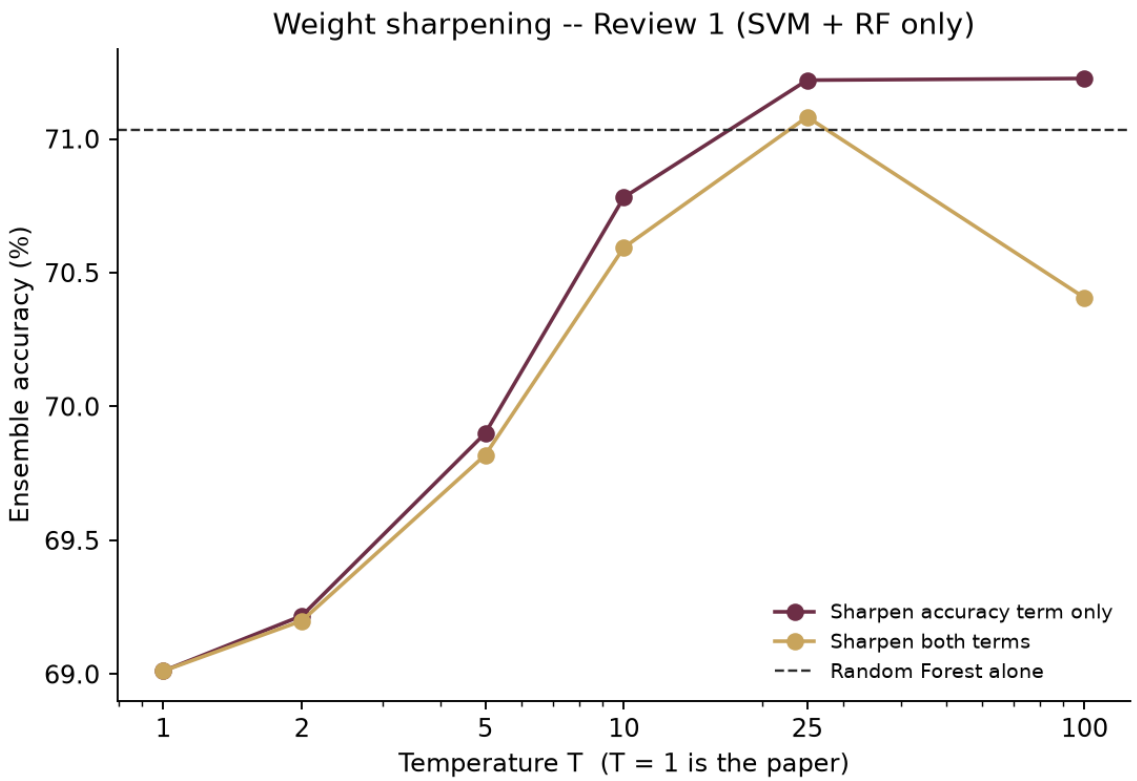
Stronger members

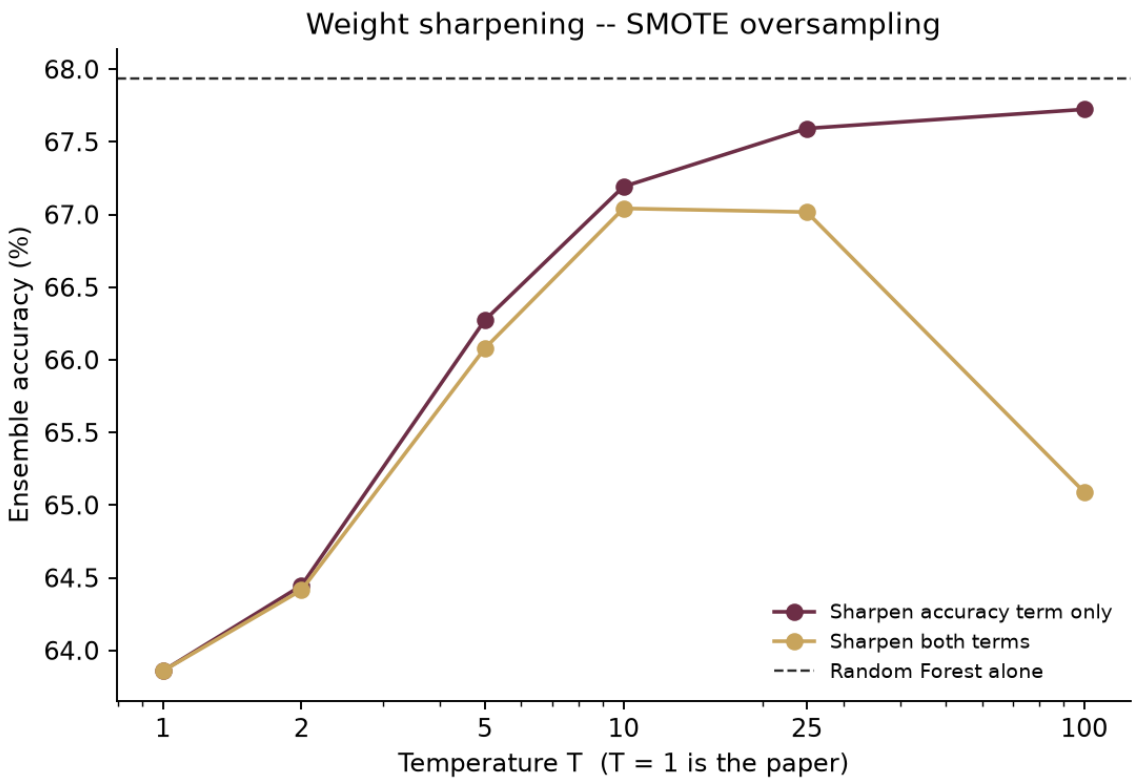
Add ExtraTrees and HistGradientBoosting, alone and combined with RF and SVM.

Tests whether better members help more than better weighting.

Route 1 — testing the Review-1 prediction

Sharpening the paper's weights with a temperature T (T = 1 is the paper exactly)





Prediction holds: sharpening the accuracy term overtakes Random Forest from T = 25 (71.22% vs 71.03%) — a small margin, and only a diagnostic.

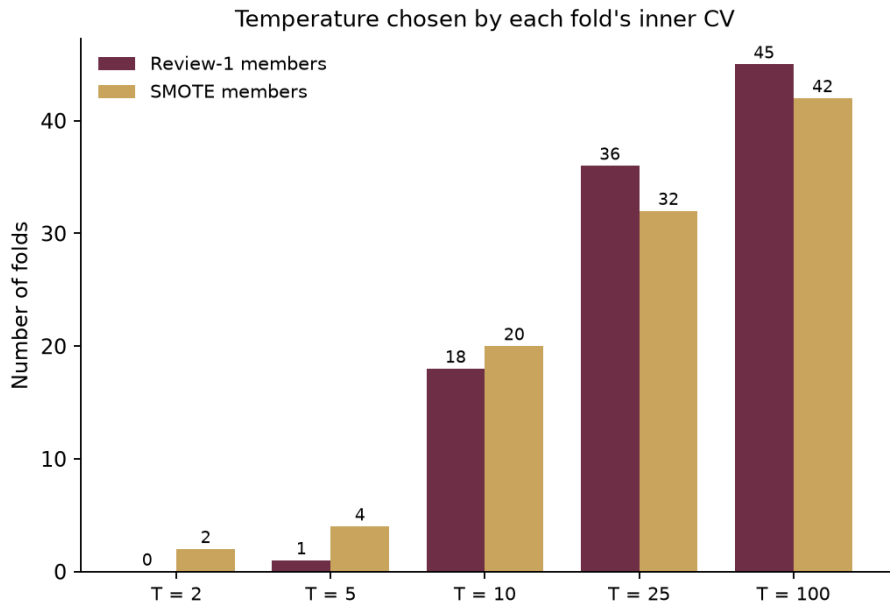
Sharpening both terms peaks at T = 25, then falls to 70.41% at T = 100: the diversity weight concentrates on the learner whose mistakes differ most — usually the weakest one.

Route 2 — the fix, done properly

Temperature chosen inside each training fold by an inner 5-fold CV — never sees the test fold

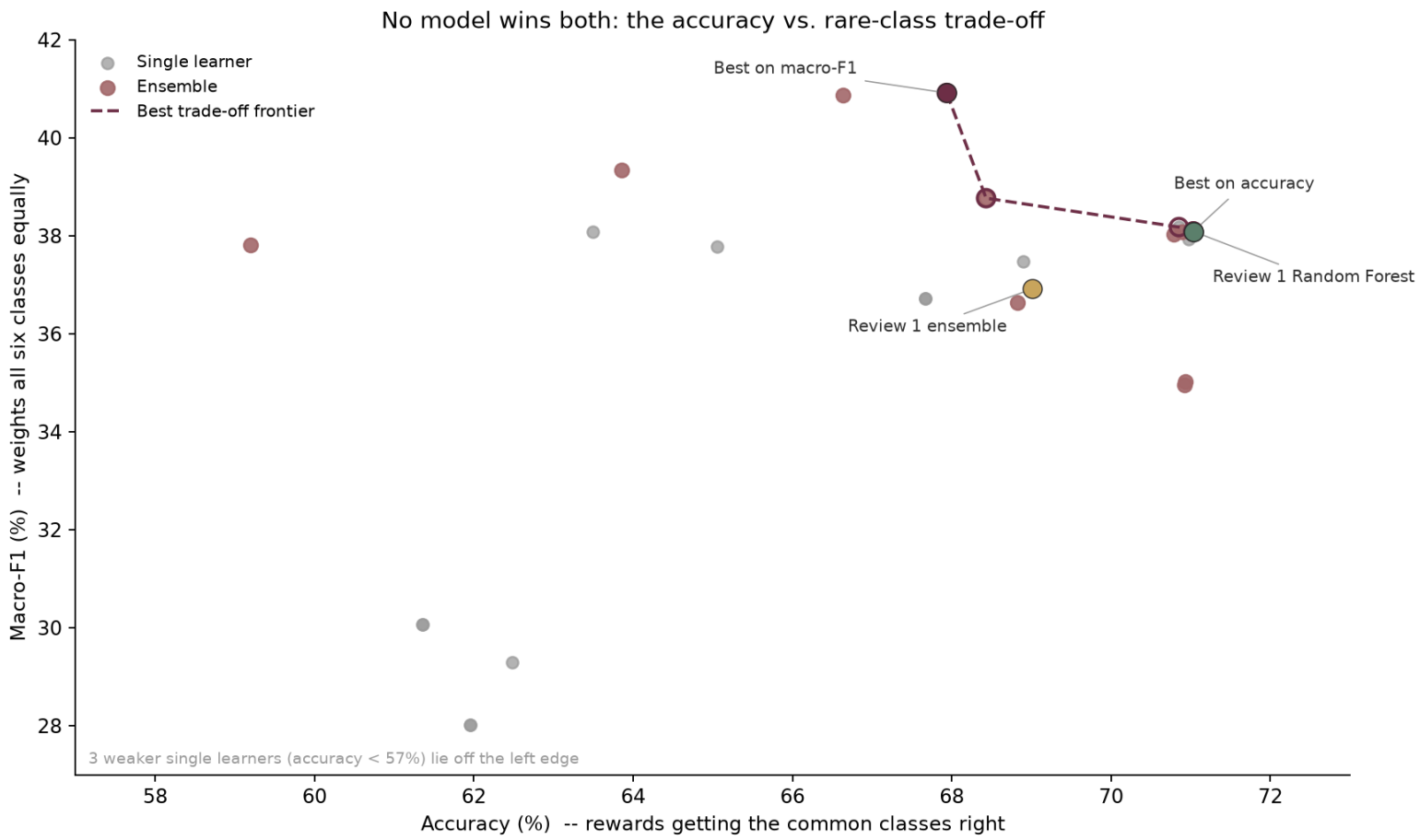
69.01%	70.89%	71.03%	99%
REVIEW-1 ENSEMBLE (T = 1)	LEARNED-T ENSEMBLE, SAME MEMBERS	RANDOM FOREST ALONE	OF 100 FOLDS CHOSE T ≥ 10

- Same four learners as Review 1, only the weighting changes: accuracy 69.01% → 70.89% (+1.88), macro-F1 36.92 → 38.08
- It closes 1.88 of the 2.02-point gap to Random Forest — now a statistical tie with it (4 wins, 2 ties, 4 losses across repeats)
- Not noise: it beats the Review-1 ensemble on accuracy in 10 of 10 repeats, on identical folds
- The inner CV, left to itself, picks a sharp temperature (T ≥ 10) in 99% of folds — independent confirmation that the paper's T = 1 is too flat
- With SMOTE members as well: macro-F1 40.87, accuracy 66.63% — the class-balanced version of the same fix



Every candidate — no model wins both metrics

28 models and ensembles, each pooled over 100 fold-runs



Reading the chart

- Right = better on the common classes; up = better on all six equally
- The dashed frontier: models no other model beats on both at once
- Best macro-F1: RF · SMOTE (40.92 / 67.94%) — tied with Learned-T ensemble · SMOTE (40.87)
- Best accuracy: RF · Review-1 setup (71.03%)
- Plain stacking reaches 70.92% accuracy but drops to 34.96 macro-F1 — the learned combiner relearns the majority bias

Review 1 → Review 2

Best model on the primary metric: RF · SMOTE

36.92 → 40.92

MACRO-F1 (PRIMARY)

35.52 → 42.07

BALANCED ACCURACY

69.01 → 67.94

ACCURACY

Metric (%)	Review-1 ensemble	Review-1 Random Forest	Review-2 best	Change
Accuracy	69.01	71.03	67.94	−1.08
Balanced accuracy	35.52	36.40	42.07	+6.55
Macro-F1	**36.92**	**38.09**	**40.92**	**+4.00**
Recall, class 3	0.00	0.00	4.00	+4.00
Recall, class 4	0.57	0.00	21.89	+21.32
Recall, class 5	81.66	81.97	78.00	−3.66
Recall, class 6	69.87	74.29	64.19	−5.69
Recall, class 7	49.90	51.06	65.43	+15.53
Recall, class 8	11.11	11.11	18.89	+7.78

If the goal is accuracy

The best-accuracy model is RF · Review-1 setup at 71.03% — +2.02 over the Review-1 ensemble. The learned-temperature ensemble gives 70.89% with the paper's own members. Neither comes near the paper's 89.73% — consistent with the Review-1 finding.

Consistent across repeats: higher macro-F1 than the Review-1 ensemble in 10 of 10 repeats (mean +3.97 points; Wilcoxon p 0.002 — indicative, as repeats share the same wines).

Feature importance — three views, checked against each other

No single importance measure is trustworthy alone

1

Permutation importance

PRIMARY

Shuffle one lab measurement in the held-out test fold and measure how much macro-F1 falls. Model-agnostic, and computed on data the model never trained on — so it measures what the model actually relies on.

2

Impurity importance (MDI)

COMPARISON

Comes free with Random Forest, but is known to favour continuous, many-valued features. Shown only to check the primary measure.

3

Direction of effect

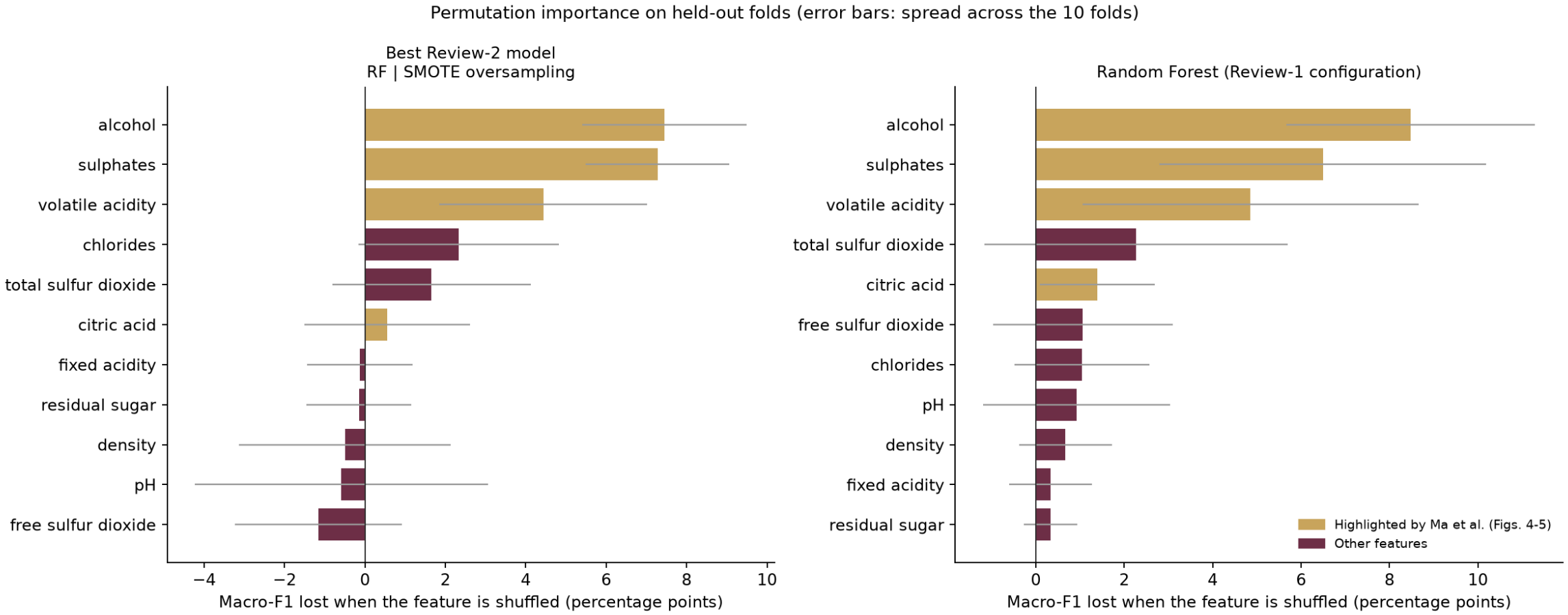
CONTEXT

Spearman correlation of each feature with quality. Importance says how much a feature matters; this says which way it pushes.

Then a practical test: retrain with only the top-k measurements, chosen inside each training fold. Choosing them on the full dataset first would leak test information into the choice.

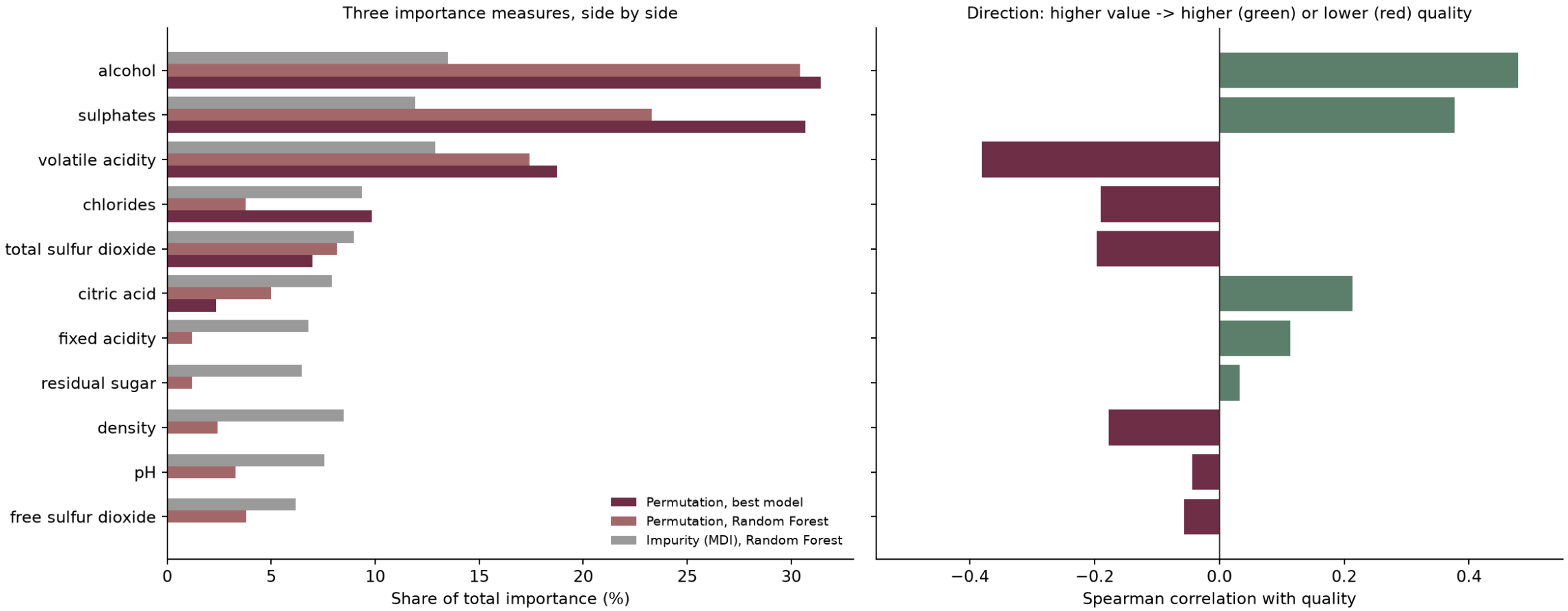
What drives the prediction

Permutation importance on held-out folds — gold bars are the attributes Ma et al. highlight



Top four: alcohol, sulphates, volatile acidity, chlorides. 3 of the paper's 4 key attributes are in our top four — citric acid ranks #6.

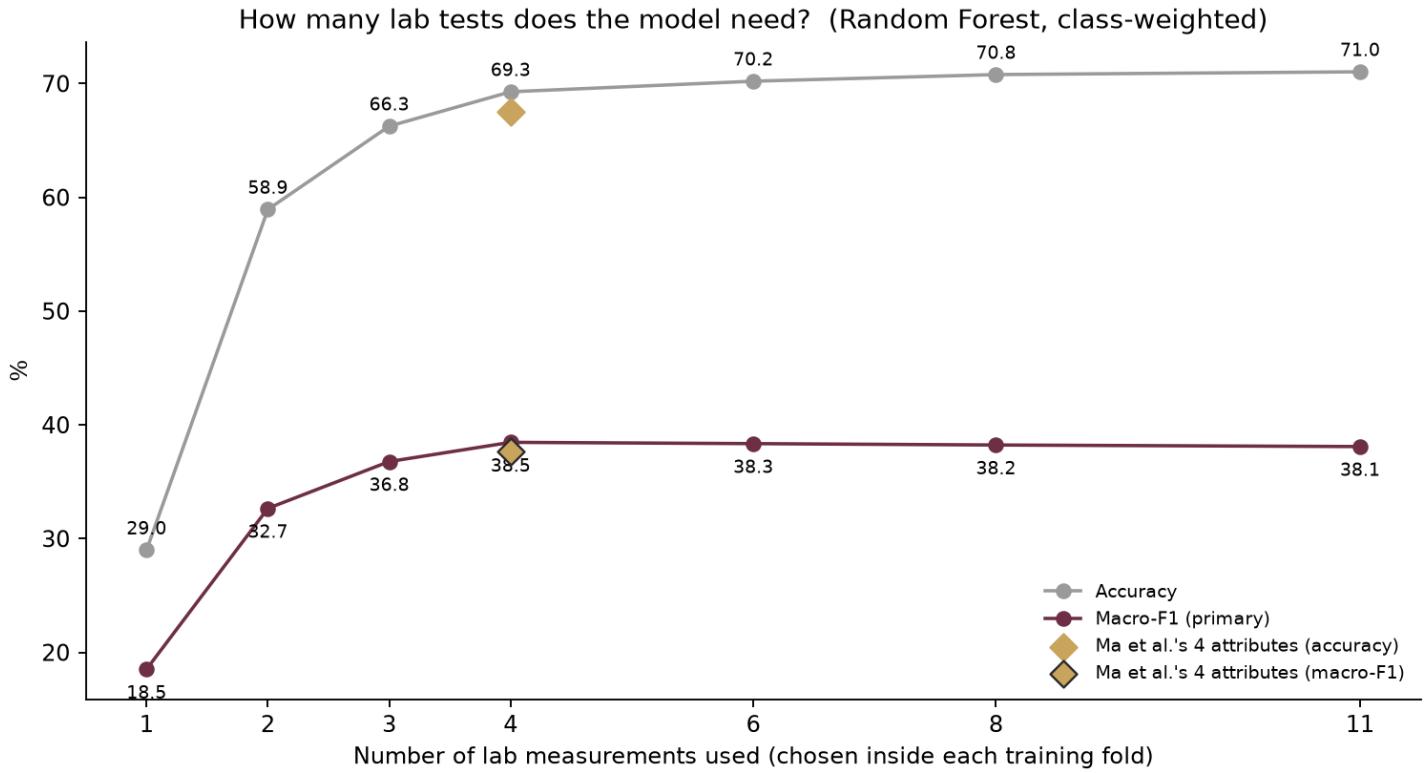
Do the methods agree — and which way does each feature push?



- Rank agreement between the three methods: 0.74 – 0.89 (1 = identical order) — the ranking is not an artefact of one method
- Better wine: higher alcohol, sulphates, citric acid. Worse wine: higher volatile acidity.

How many lab tests does the model need?

Random Forest, class-weighted; features chosen inside each training fold



Business reading

- 4 tests (volatile acidity; sulphates; alcohol; chlorides) reach 69.27% accuracy — 98% of what all 11 achieve (71.03%)
- Macro-F1 with 4 tests: 38.46 vs 38.09 with all 11
- The paper's own 4 attributes: 67.55% — below our data-chosen four
- A winery screening batches could run a much smaller test panel for most of the predictive value

Conclusion

Class balancing

Done. SMOTE oversampling lifts macro-F1 from 36.92 to 39.34 and rare-class recall sharply, at a stated cost in accuracy.

Improving the result

The Review-1 diagnosis was tested and confirmed, then fixed: a temperature learned inside each fold takes the paper's own ensemble from 69.01% to 70.89%. Best macro-F1 overall: 40.92.

Feature importance

Alcohol, sulphates, volatile acidity drive the prediction; three methods agree; 3 of the paper's 4 key attributes confirmed. Four lab tests carry most of the value.

What did not change

No configuration comes near the paper's 89.73% — the best accuracy here is 71.03%. Review 2 strengthens the Review-1 conclusion rather than overturning it.

Limitations, future work and references

Limitations

- The best model is chosen on the same CV that scores it — a small optimistic bias
- The 10 repeats reuse the same wines, so the paired test across them is indicative, not a formal confidence interval
- Class 3 has 10 wines in total; no method can learn it reliably from that
- Untuned hyperparameters throughout — kept fixed so comparisons stay fair

Future work

- Nested CV around the model choice itself, and a corrected resampled t-test for proper confidence intervals
- Ordinal models that know a 7 is closer to an 8 than to a 3
- Repeat everything on the 4,898-sample white wine set

References

[1] Ma, R., Mao, D., Cao, D., Luo, S., Gupta, S., & Wang, Y. (2024). From vineyard to table: Uncovering wine quality for sales management through machine learning. Journal of Business Research, 176, 114576.

[2] Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. Decision Support Systems, 47(4), 547–553.

[3] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research, 16, 321–357.

[4] Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

[5] Wolpert, D. H. (1992). Stacked generalization. Neural Networks, 5(2), 241–259.

[6] Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. JMLR, 12, 2825–2830.

[7] UCI Machine Learning Repository — Wine Quality Dataset.

Thank you — questions welcome.

Backup 1 — every candidate, ranked by macro-F1

Pooled over 100 fold-runs

#	Model	Kind	Accuracy	Bal. acc.	Macro-F1
1	RF · SMOTE	single learner	67.94	42.07	40.92
2	Learned-T ensemble · SMOTE	ensemble	66.63	42.49	40.87
3	Paper ensemble · SMOTE	ensemble	63.86	41.44	39.34
4	Paper ensemble · class weights	ensemble	68.43	37.94	38.77
5	RF · no balancing	single learner	70.85	36.70	38.18
6	RF · Review-1 setup	single learner	71.03	36.40	38.09
7	RF · class weights	single learner	71.03	36.40	38.09
8	GradBoost · SMOTE	single learner	63.49	39.40	38.08
9	Learned-T ensemble · Review-1 setup	ensemble	70.89	36.58	38.08
10	Paper weights: SVM + RF + ExtraTrees + HistGB	ensemble	70.79	36.96	38.04
11	ExtraTrees	single learner	70.97	36.45	37.93
12	Stacked: paper's 4 learners, balanced combiner	ensemble	59.20	50.03	37.82
13	GradBoost · class weights	single learner	65.05	37.92	37.78
14	HistGB	single learner	68.89	36.86	37.48

Showing the top 14 of 28. Full table: results/review2/tables/model_comparison.csv

Backup 2 — confusion matrices, Review 1 vs Review 2

Pooled over 100 fold-runs (15,990 predictions); rows = true class

Review-1 ensemble

true \ pred	3	4	5	6	7	8
3	0	9	70	21	0	0
4	1	3	372	144	10	0
5	4	11	5561	1202	31	1
6	3	17	1533	4458	360	9
7	0	0	114	874	993	9
8	0	0	0	96	64	20

Review-2 best: RF · SMOTE

true \ pred	3	4	5	6	7	8
3	4	39	46	11	0	0
4	29	116	253	111	21	0
5	62	280	5312	1040	110	6
6	18	162	1223	4095	756	126
7	3	18	81	511	1302	75
8	0	0	0	40	106	34

The diagonal is correct predictions. Balancing moves predictions off the 5/6 columns into the rare classes — more rare wines caught, and some common wines now mislabelled as rare.

Backup 3 — weight-sharpening sweep (diagnostic)

Review-1 members; ensemble accuracy and largest average weight at each temperature

T	Accuracy — accuracy term only	Largest weight	Accuracy — both terms	Largest weight
1	69.01	0.270	69.01	0.270
2	69.22	0.293	69.20	0.291
5	69.90	0.364	69.82	0.356
10	70.78	0.466	70.59	0.448
25	71.22	0.590	71.08	0.546
100	71.23	0.623	70.41	0.523

Random Forest alone: 71.03%. T = 1 is the paper's own scheme. Never used to pick the final model — the weights here are computed from the predictions they are scored on.

Backup 4 — feature-importance table

Percentage points of macro-F1 lost when the feature is shuffled; Spearman ρ with quality

Rank	Feature	Best model	Random Forest	RF impurity	ρ with quality	Paper key?
1	alcohol	7.45	8.48	0.135	0.48	yes
2	sulphates	7.28	6.49	0.119	0.38	yes
3	volatile acidity	4.44	4.86	0.129	-0.38	yes
4	chlorides	2.33	1.05	0.093	-0.19	
5	total sulfur dioxide	1.66	2.27	0.089	-0.20	
6	citric acid	0.56	1.39	0.079	0.21	yes
7	fixed acidity	-0.13	0.34	0.068	0.11	
8	residual sugar	-0.15	0.34	0.065	0.03	
9	density	-0.50	0.67	0.085	-0.18	
10	pH	-0.59	0.92	0.076	-0.04	
11	free sulfur dioxide	-1.16	1.06	0.062	-0.06	

Backup 5 — is the improvement consistent?

Paired comparison across the 10 repeats — every model scored on identical folds

Model	Compared with	Metric	Mean diff.	Repeats won	Wilcoxon p
RF · SMOTE	Paper ensemble · Review-1 setup	Macro-F1	+3.97	10 / 10	0.002
RF · SMOTE	Paper ensemble · Review-1 setup	Accuracy	−1.08	1 / 10	0.004
RF · SMOTE	RF · Review-1 setup	Macro-F1	+2.80	10 / 10	0.002
RF · SMOTE	RF · Review-1 setup	Accuracy	−3.10	0 / 10	0.002
RF · SMOTE	Learned-T ensemble · SMOTE	Macro-F1	+0.01	4 / 10	0.846
RF · SMOTE	Learned-T ensemble · SMOTE	Accuracy	+1.31	9 / 10	0.006
Learned-T ensemble · Review-1 setup	Paper ensemble · Review-1 setup	Macro-F1	+1.16	10 / 10	0.002
Learned-T ensemble · Review-1 setup	Paper ensemble · Review-1 setup	Accuracy	+1.88	10 / 10	0.002
Learned-T ensemble · Review-1 setup	RF · Review-1 setup	Macro-F1	−0.01	6 / 10	0.922
Learned-T ensemble · Review-1 setup	RF · Review-1 setup	Accuracy	−0.14	4 / 10	0.438
RF · Review-1 setup	Paper ensemble · Review-1 setup	Macro-F1	+1.17	10 / 10	0.002
RF · Review-1 setup	Paper ensemble · Review-1 setup	Accuracy	+2.02	10 / 10	0.002

Caveat: the repeats reuse the same 1,599 wines with different splits, so they are not independent samples and the p-values are optimistic. The win count is the plainer evidence.

Backup 6 — the Review-1 finding, for reference

Why the paper's 89.73% does not reproduce — three checks from its own tables

1 Protocol mismatch

The paper's Table 9 confusion matrix sums to 491 samples — 30.7% of 1,599 — a single hold-out split, not the 100-run CV the text describes.

2 Internal arithmetic

The diagonal of that table is 403 of 491 = 82.08%, 7.65 points below the 89.73% stated in the text.

3 Flat weights

Table 8's weights are never more than 0.053 from 0.25; applied to the paper's own base accuracies they give 67.52%.

Caveat: the Table 9 cells were transcribed by eye from the printed PDF. Reported as "we could not reconcile these figures" — not as a claim that the paper is wrong.